Sketching When an AI Tutor Should Abstain: A Study Protocol for Concept-Based Design of Abstention in Retrieval-Augmented Question Answering

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Study protocol, version 2 (November 12, 2026). First posted October 8, 2026. No results yet. Not peer reviewed.

What changed in version 2: the study had not started by the original dates, so the schedule now begins November 16, 2026. I also revised the wording and checked every reference against publisher records. The research questions, method, and analysis plan are unchanged.

Abstract

In earlier work I built a retrieval-augmented question-answering system from standard components and found that it answered all 258 unanswerable questions in a 500-question SQuAD 2.0 sample, in the same form as its correct answers. A tutor that never declines to answer gives learners no cue for when to doubt it. Abstention is usually treated as a scoring problem: a model or calibrator outputs one confidence value, and the system answers when that value is high enough. This protocol describes a study that treats abstention as a design problem instead. Using model sketching, a method for building quick functional models out of concepts people can read, I will write concepts that describe why a tutor should decline (for example, "the retrieved passage does not mention the question's main entity"), combine them into small sketch models, and compare those with a retrieval-score threshold on the same 500 questions and on a held-out sample of 200. A dated design journal will record how sketching changes the way I frame the problem. The study reuses existing data and involves no participants. I am posting the research questions, procedure, analysis plan, limitations, and timeline before collecting any results.

Keywords: abstention; model sketching; retrieval-augmented generation; unanswerable questions; trust calibration; AI tutors

1. Introduction

Students increasingly ask AI systems questions they cannot yet answer themselves. In that situation, the only way a learner can calibrate trust is if the system gives some signal of when it has no basis for an answer. When I studied a retrieval-augmented question-answering (RAG) system, that signal was missing entirely. The system answered every unanswerable question it was given (Choi, 2026).

The usual fix is a better score. A calibrator estimates the probability that an answer is correct, and the system abstains below a threshold (Kamath et al., 2020). That can work. What bothers me about it is that a single number hides *why* the system should not answer, and the reason is exactly what an honest tutor could tell a learner. So in this study I want to try designing the reasons for abstaining directly, as named

concepts, using model sketching (Lam et al., 2023). My guiding question is whether naming those reasons shows anything a single confidence score hides.

I am writing this as a protocol so that the research questions, procedure, and analysis are fixed before any sketching starts, and the eventual report can be checked against what I planned.

2. Background

2.1 Abstention in question answering

SQuAD 2.0 added over 50,000 unanswerable questions, written by crowdworkers to look similar to answerable ones, so that systems must "determine when no answer is supported by the paragraph and abstain from answering" (Rajpurkar et al., 2018). Kamath et al. (2020) studied selective question answering under domain shift. They found that abstention policies based only on a model's softmax probabilities do poorly, because models are overconfident on out-of-domain inputs, while a separately trained calibrator did better: at 80% accuracy it answered 56% of questions, compared with 48% when using the model's probabilities directly.

2.2 Model sketching

Lam et al. (2023) observed that machine learning practitioners often get stuck on low-level technical details such as architectures and metrics. They introduced model sketching, in which practitioners compose high-level, human-understandable concepts into quick functional approximations of a model. Their Python package, ModelSketchBook, scores each concept zero-shot (for text, by prompting GPT-3) and combines the concept scores with simple aggregators such as logistic regression or a decision tree. In an evaluation with 17 practitioners, sketching moved attention from implementation to higher-level exploration and helped participants find gaps in how they had formulated the problem.

2.3 The gap

Each of these lines of work leaves something open for my purposes. Rajpurkar et al. (2018) give a benchmark with unanswerable questions, not a way to design abstention. Kamath et al. (2020) train a calibrator that decides when to abstain, but its decision is one score rather than reasons a person can inspect. Dense passage retrieval (Karpukhin et al., 2020), which my pipeline uses, finds passages, but finding a passage is not the same as having a basis to answer. And model sketching has been evaluated on tasks like content moderation, not on abstention or education. I have not found work that uses model sketching to design when a learner-facing system should decline to answer.

3. Research Questions

- **RQ1.** Which human-understandable concepts separate the questions the system should have declined from those it should have answered?
- **RQ2.** How do concept-based sketch models of abstention compare with a single retrieval-score threshold on the same questions?
- **RQ3.** What does the sketching process reveal about the abstention problem that the original metrics (exact match, F1, Recall@K) did not?

4. Method

4.1 Data and system

The study reuses the evaluation run from my earlier work, and no new data are collected. The question set is a fixed, seeded sample of 500 questions from the SQuAD 2.0 validation split: 242 answerable and 258 labeled unanswerable. The retriever is Dense Passage Retrieval (Karpukhin et al., 2020) with the publicly released Natural Questions checkpoints, searched through an exact inner-product FAISS index over about 29,000 chunks drawn from about 20,000 SQuAD 2.0 passages. The generator is Flan-T5-base, prompted with no instruction about unanswerable questions. For each question I need the question itself, the top retrieved passages and their scores, the generated answer, and the gold label. If I find that the per-question outputs were not saved, I will rerun the pipeline with the same seed before starting.

4.2 Procedure

1. **Baseline.** Fit a single threshold on the top-1 retrieval score to predict "should abstain" (the 258 unanswerable questions), and record accuracy, precision, recall, and F1.
2. **Concept authoring.** Write 6 to 10 concepts as short natural-language prompts. My starting candidates are: the retrieved passage does not mention the question's main entity; the question asks about something the passage negates; the question contains a false premise; the top passages disagree with each other; and answering would need information that is not in the passage.
3. **Sketching.** Combine concepts into 3 to 5 sketch models (single concepts, small sets, all concepts). After each sketch, read 20 misclassified questions and revise or add concepts.
4. **Comparison.** Evaluate every sketch and the baseline on the same 500 questions, then check the best sketch once on a fresh seeded sample of 200 SQuAD 2.0 validation questions.

One practical risk: ModelSketchBook was built on GPT-3, which is no longer offered. Before starting, I will check whether the package runs with a current model and estimate how many model calls I will need (concepts times questions).

4.3 Design journal

During steps 2 and 3 I will keep a dated design journal, noting each concept I add, change, or drop, and why. The journal is the data for RQ3.

5. Analysis Plan

- **RQ1:** precision and recall of each concept against the abstain label, with example questions each concept catches and misses.
- **RQ2:** abstention F1 for each sketch and the baseline, with bootstrap 95% intervals over the 500 questions. The held-out 200 questions guard against tuning to the sample.
- **RQ3:** qualitative coding of journal entries for moments when sketching changed how I framed the problem, such as a new concept, a dropped concept, or a relabeled question.

6. Limitations

I am the only designer, so RQ3 will be a reflective design account, not a user study. SQuAD questions are not student questions, and SQuAD passages are not course materials. Zero-shot concept scores depend on

the underlying model and may not carry over to another one. Finally, no learners are involved, so anything I say about learners' trust will be an implication, not a finding.

7. Timeline

Table 1. Planned schedule (2026–2027). No part of the study had started when this revision was posted.

Stage	Dates
Check saved outputs and package compatibility	Nov 16 to Nov 22, 2026
Baseline and concept authoring	Nov 23 to Dec 6, 2026
Sketching and iteration	Dec 7 to Dec 20, 2026
Held-out check and analysis	Dec 21, 2026 to Jan 3, 2027
Writing the report	Jan 4 to Jan 17, 2027
Posting the report	Jan 18, 2027

8. Declarations

Origin of this work. This study follows up on my final project for CSCI E-222, Foundations of Large Language Models, at Harvard University (Spring 2026), reported in Choi (2026).

Use of AI tools. Claude (Anthropic) helped draft the decision, revise the wording.

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Competing interests. None.

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